

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	02 October 2026
Team ID	SWTID-2026-1268
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Technical Architecture:

The architecture below shows how a record request passes through a script-controlled ACL on the ServiceNow platform. The ACL script reads the record's field value, compares it with the user's roles, and returns allow or deny. The local user layer, the cloud instance, and external interfaces (SSO/LDAP) are shown separately.

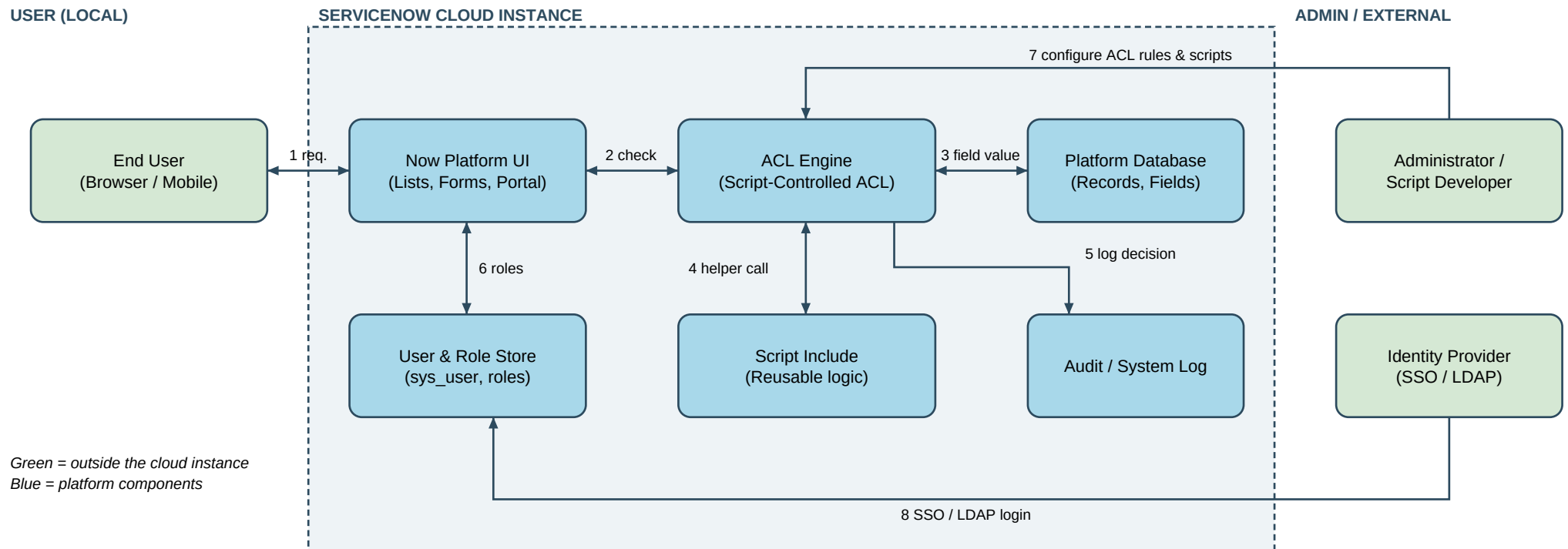


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How the user interacts with the application: browsing lists and opening record forms	ServiceNow Next Experience UI / Service Portal, HTML, CSS, JavaScript, AngularJS
2.	Application Logic-1	ACL rule definition: table, field, operation and required role	ServiceNow Access Control (ACL) rules, role-based access
3.	Application Logic-2	ACL script that reads the record's field value and the current user, and returns true (allow) or false (deny)	Server-side JavaScript (GlideRecord, GlideSystem, current / answer variables)
4.	Application Logic-3	Reusable helper functions shared by several ACL scripts	Script Include (JavaScript)
5.	Database	Records, users and roles stored in related tables with field-level data	ServiceNow relational database (Now Platform tables)
6.	Cloud Database	Database service hosted and managed with the cloud instance	ServiceNow-managed cloud database
7.	File Storage	Attachments on records, protected by the same ACLs	ServiceNow attachment storage (sys_attachment)
8.	External API-1	Single Sign-On / directory login for user identity	SAML 2.0 SSO, LDAP
9.	External API-2	Programmatic access to records by other systems, with ACLs enforced on every call	ServiceNow REST Table API
10.	Machine Learning Model	Not used in this version (rule-based access control). Optional future use: spotting unusual access patterns	Not applicable
11.	Infrastructure (Server / Cloud)	Application deployment on cloud. Local Server Configuration: none (browser only). Cloud Server Configuration: ServiceNow cloud instance (Personal Developer Instance for development)	ServiceNow Cloud

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Open-source frameworks used on the client side and for scripting	AngularJS, Bootstrap, JavaScript (Rhino engine)
2.	Security Implementations	Deny-by-default access; role plus field-value script checks; field-level restrictions; encrypted traffic; SSO login; audit log of every decision	ACL rules, role-based access control, HTTPS/TLS, SAML SSO, OWASP guidelines
3.	Scalable Architecture	Multi-tier design (browser, application, database); lightweight scripts so more users, tables and rules can be added without redesign	Multi-tier ServiceNow architecture, Script Includes
4.	Availability	Managed cloud hosting with redundant infrastructure and failover, so access control is available whenever the platform is	ServiceNow cloud high availability, load balancing
5.	Performance	Cheap conditions evaluated before scripts; indexed fields used in conditions; no heavy queries inside ACL scripts; ACL results cached per session	Database indexes, ACL caching, condition-first evaluation